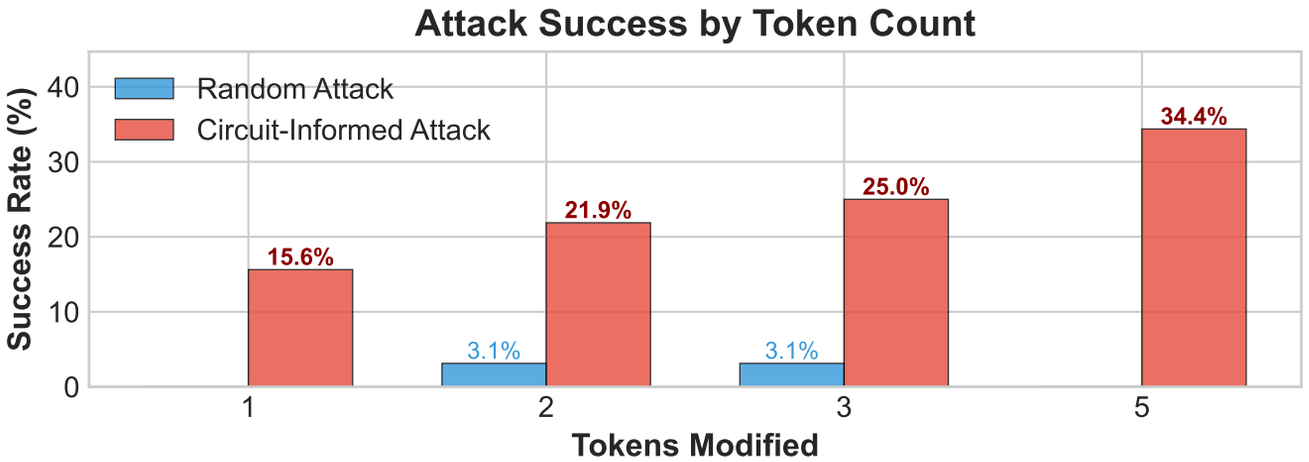
Circuit-Informed Adversarial Attacks: Detailed Analysis

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                KEY FINDINGS
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ATTACK SUCCESS RATES
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  Random Attacks:           2/128 = 1.6%
  Circuit-Informed:         31/128 = 24.2%

  Improvement:              16× more effective

EFFICIENCY (Success per Token Modified)
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  Random avg:               0.0065
  Circuit-informed avg:     0.1044

  Circuit is 16× MORE EFFICIENT

STATISTICAL SIGNIFICANCE
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  Fisher's exact test:      p = 0.000000
  Result:                   ✓ HIGHLY SIGNIFICANT (p < 0.001)

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              MAIN CONCLUSION
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  Understanding a model's internal circuits
  enables dramatically more effective
  adversarial attacks.

  This demonstrates that interpretability
  research has important security implications.

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